

AHU screening (single-system, 2026-06)

Offline pandas-oracle pack for one AHU folder or a building folder of device subfolders. This is not the BUILDING_100 Overview-mirrored lab PDF. Figures use the PyPI Plotly helpers shared with the Railway UI.

AHU_1

Executive summary

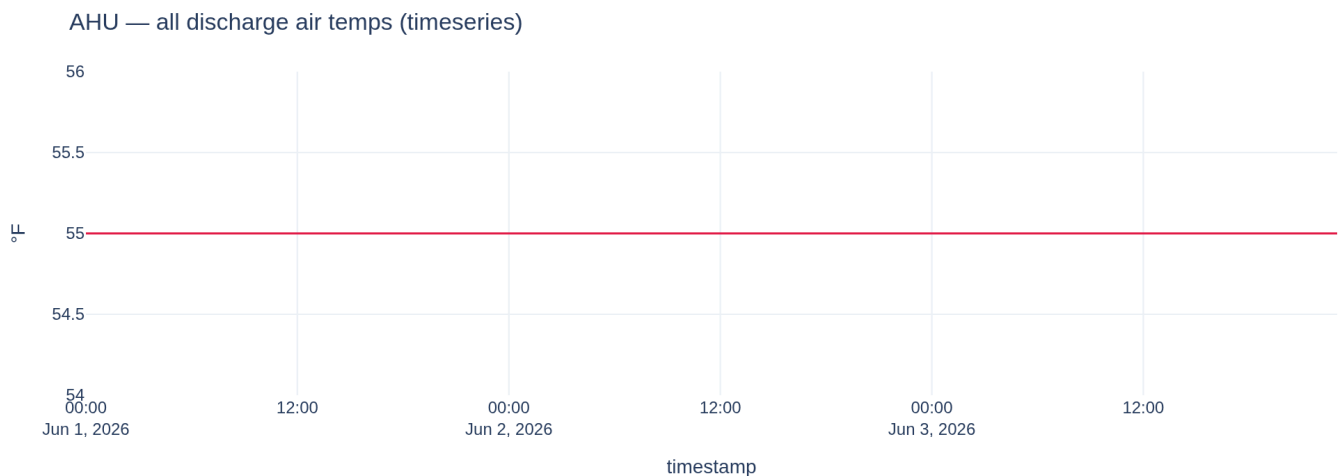
AHU_1 in 2026-06: 72 samples, 72.0 h. Data issues: outside-air-temp has 1 sample(s) outside -60..130 (min 35.0, max 200.0). Confirmed faults: SV-RANGE 1.0 h, SV-FLATLINE 71.0 h, SV-SPIKE 2.0 h, SV-RATE 2.0 h, FC1 72.0 h, ECON-1 26.0 h, SCHED-247 72.0 h. No web outdoor-air temperature was joined, so rules that require it stay skipped. Fan is ON for 100.0 percent of samples. Unsupervised anomaly minutes are not confirmed FDD faults.

RCx week (2026-06-01 to 2026-06-03 UTC)

One representative week inside the filtered month, preferring fan-ON coverage. Lines use the PyPI Plotly helpers shared with Railway RCx plots.

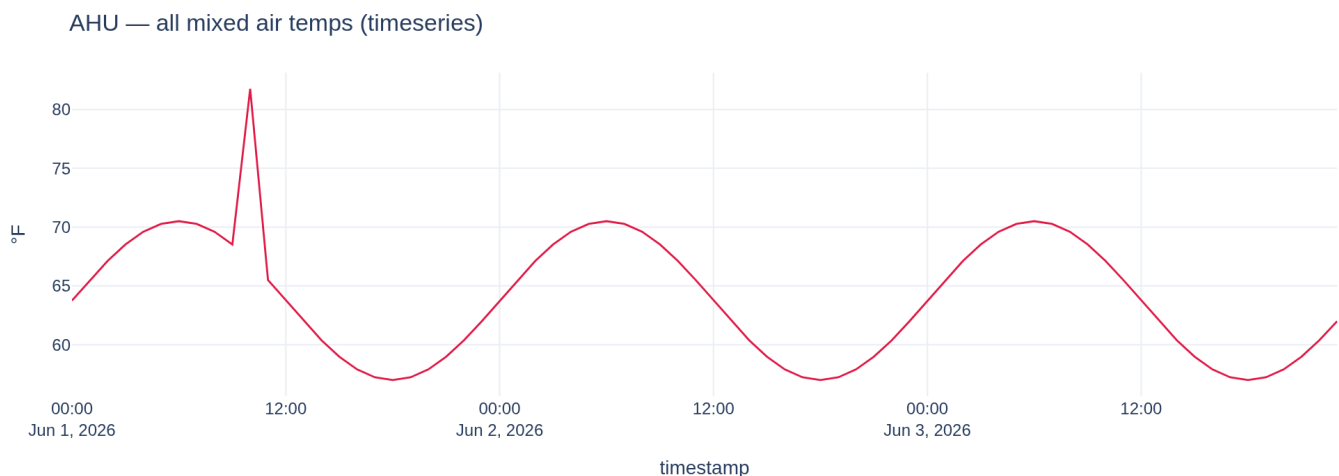
AHU — all discharge air temps (timeseries)

SAT / DAT for every AHU.

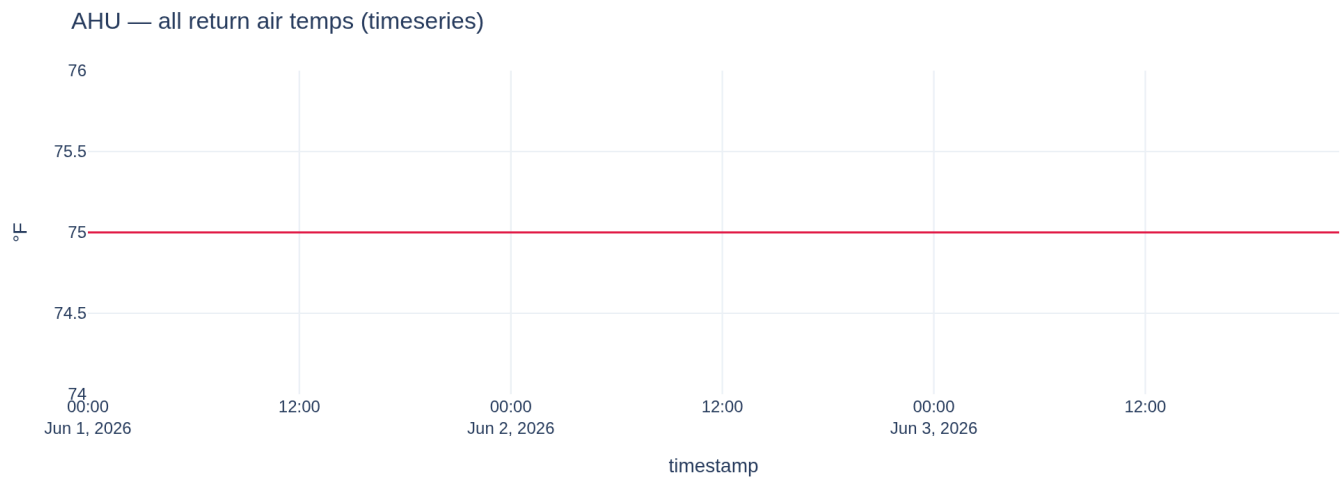


AHU — all mixed air temps (timeseries)

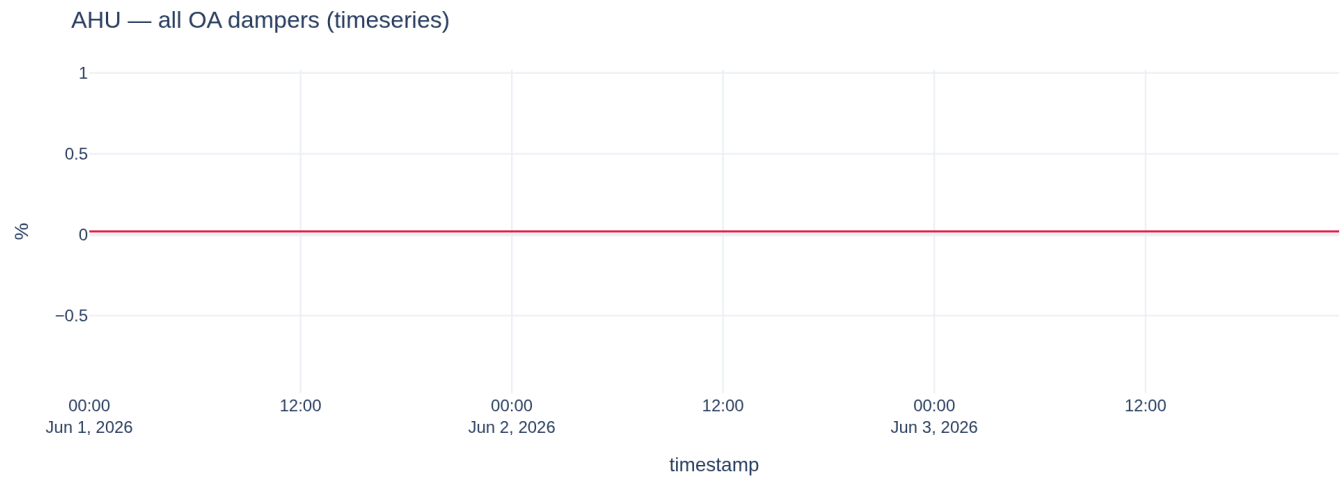
MAT across AHUs.



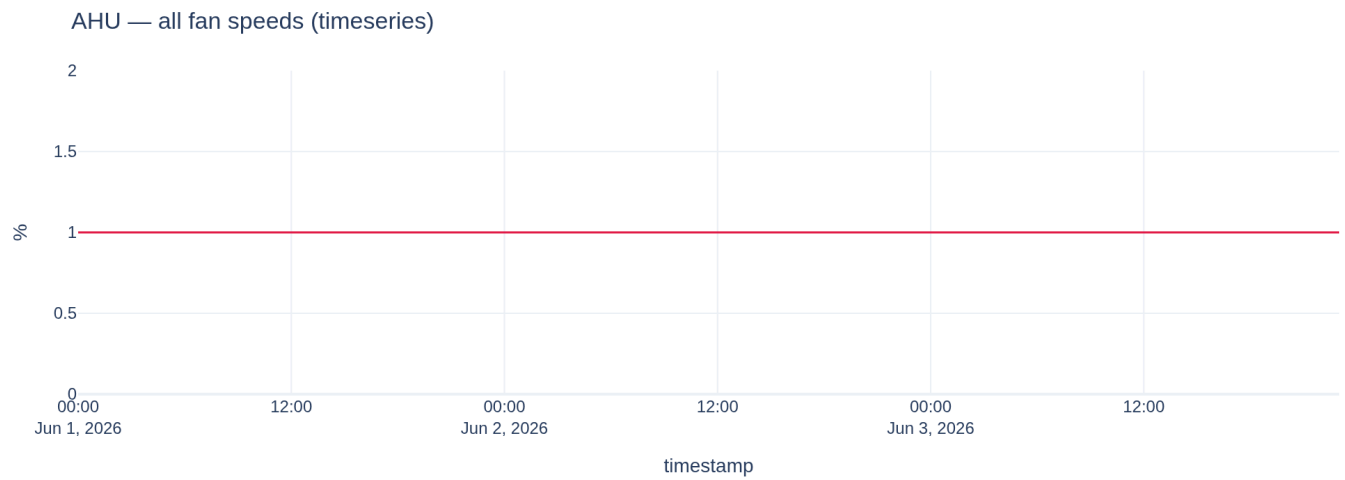
AHU – all return air temps (timeseries)
RAT across AHUs.



AHU – all OA dampers (timeseries)
OA damper % across AHUs (preferred % companion for SAT/MAT/RAT views).

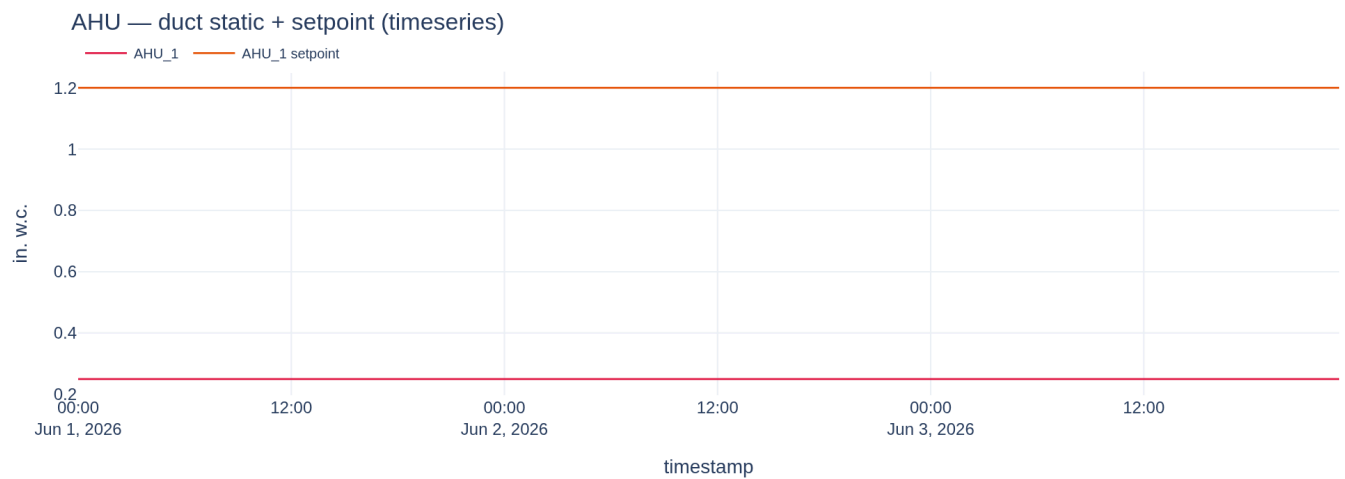


AHU – all fan speeds (timeseries)
Fan command % across AHUs – pair with duct-static / FC1 pressure views, not SAT/MAT.



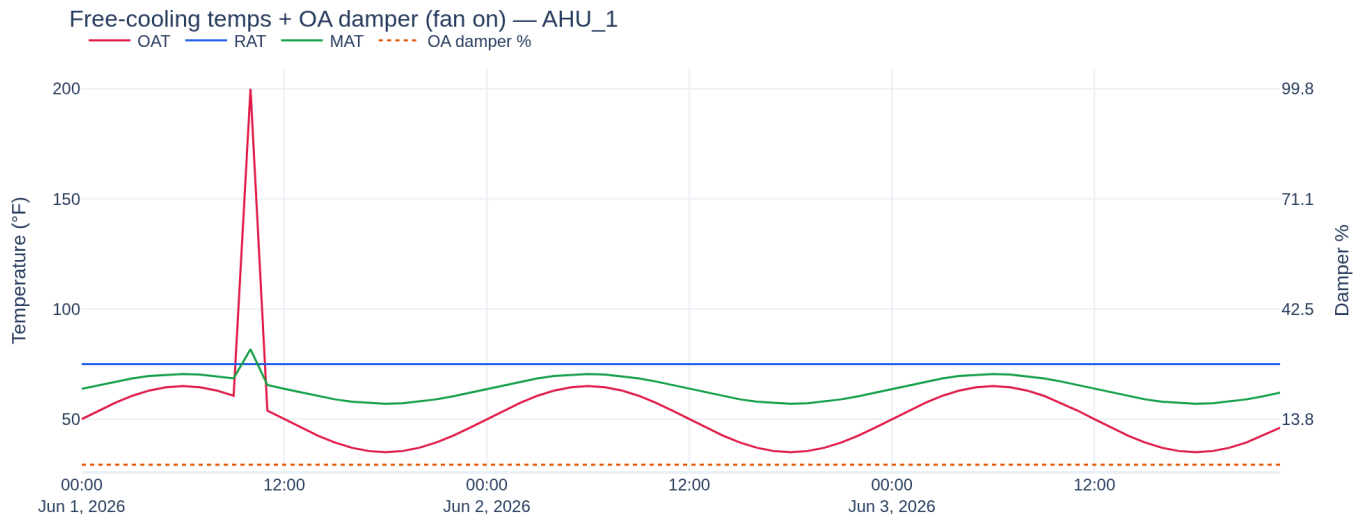
AHU — duct static + setpoint (timeseries)

Duct static pressure over time per AHU, with the mapped setpoint as a companion trace – flat setpoint + high static while fan runs means a reset opportunity.



Free-cooling temps + OA damper

OAT, RAT, MAT, and SAT on the temperature axis; OA damper percent on the right axis.



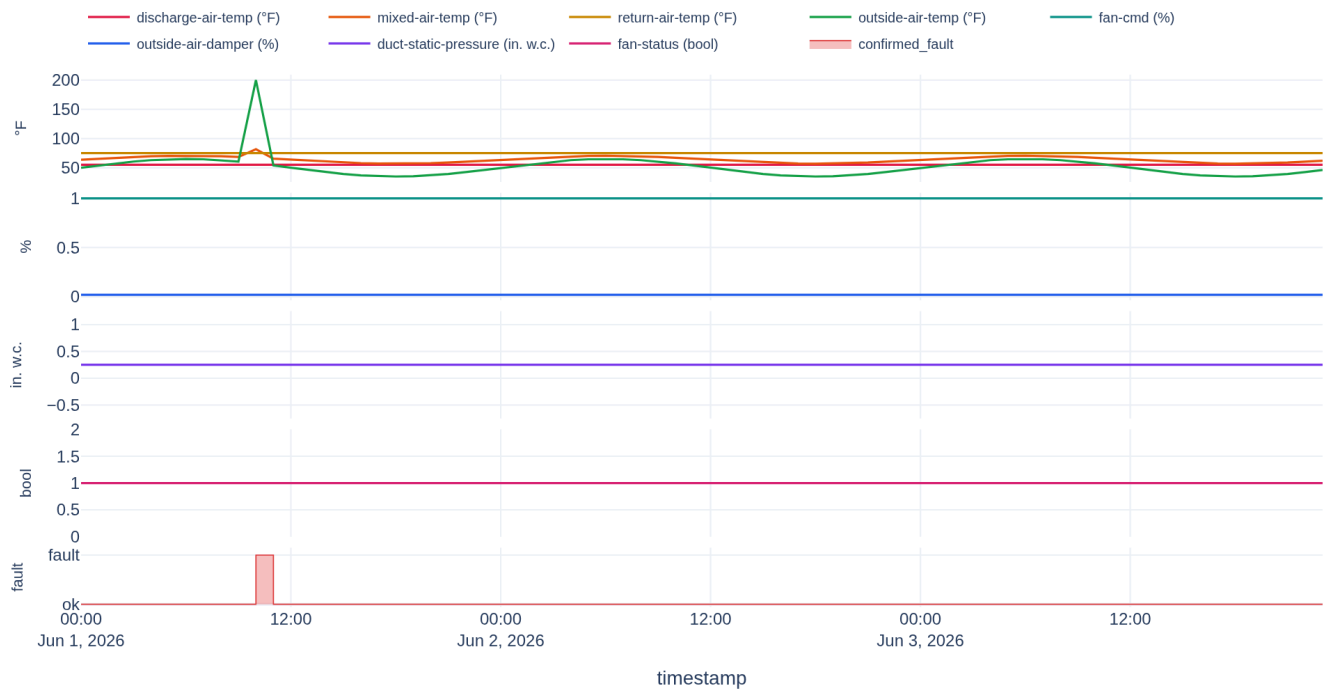
Confirmed faults

A figure is included only when the cookbook rule has confirmed fault hours in the filtered month.

SV-RANGE – Sensor out of hard range

Rule: Flags any modeled sensor reading outside its physical hard range.

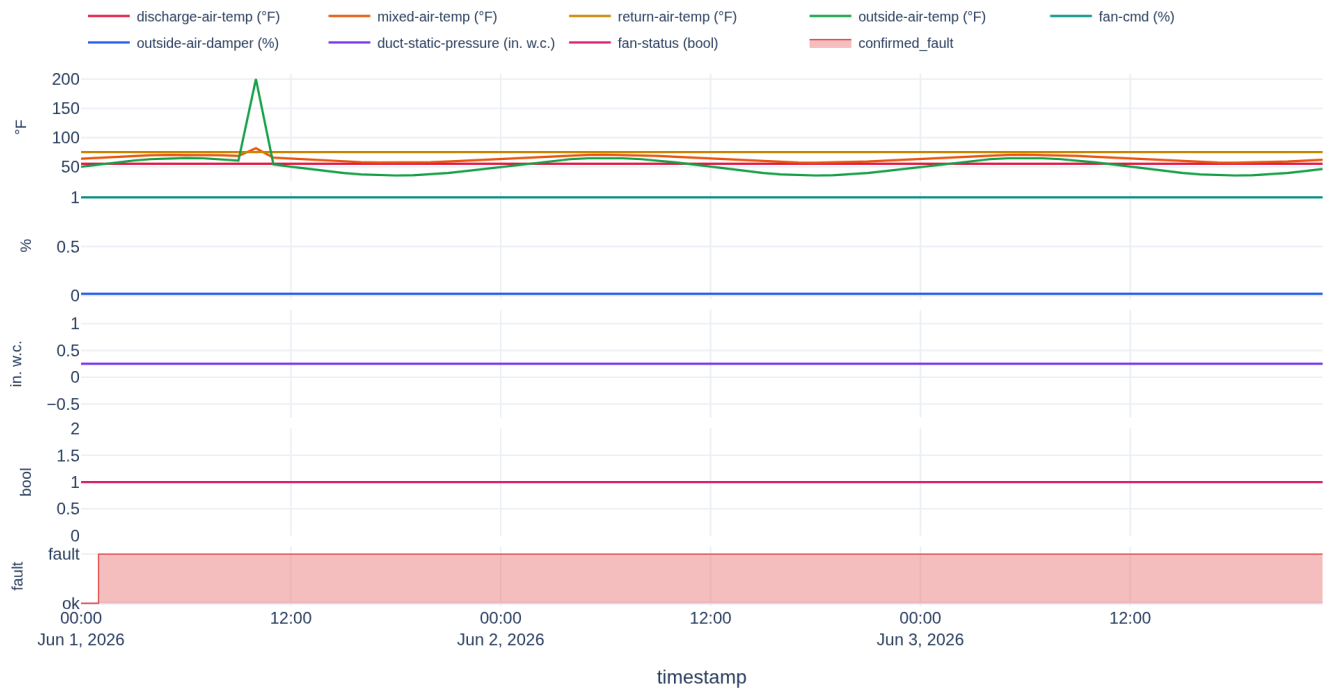
Data: 1.0 fault hours in 2026-06. 1.0h fault (1.4% of active)



SV-FLATLINE – Sensor flatline (stuck)

Rule: Flags a sensor that stays stuck within a tiny band for too long.

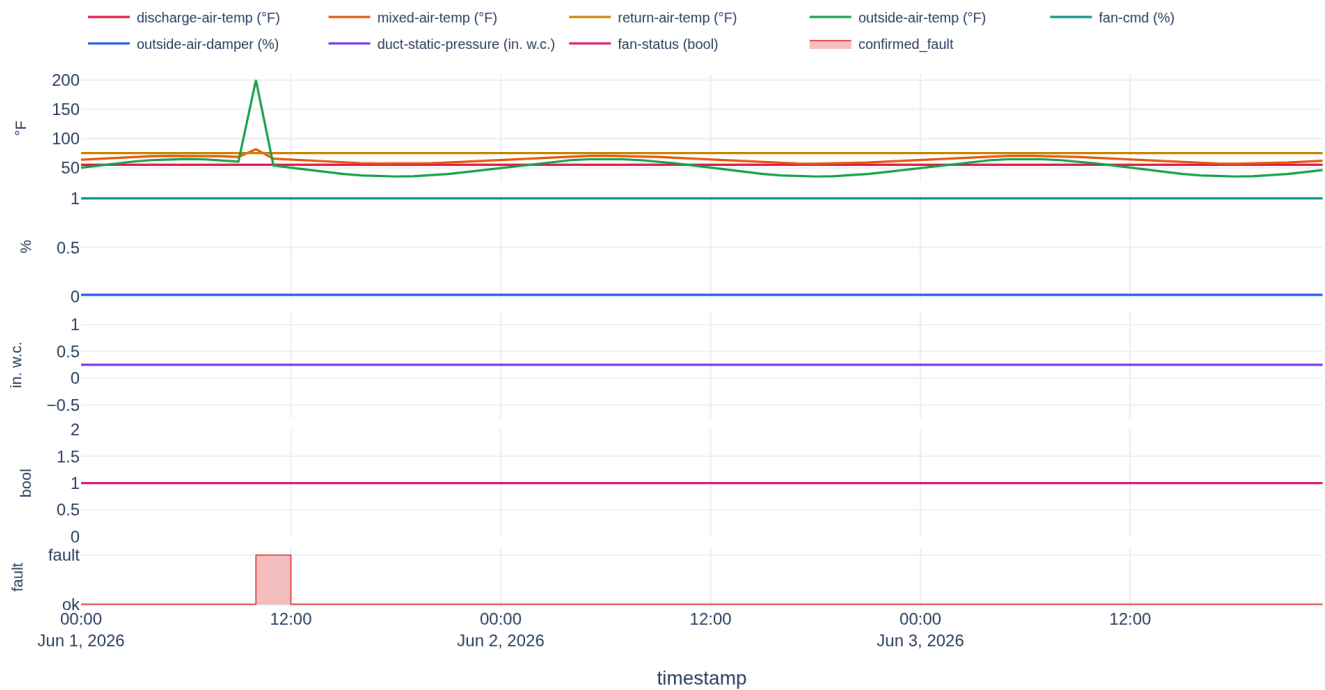
Data: 71.0 fault hours in 2026-06. 71.0h fault (98.6% of active)



SV-SPIKE – Sensor rate-of-change spike

Rule: Flags an implausible sample-to-sample jump for the sensor type.

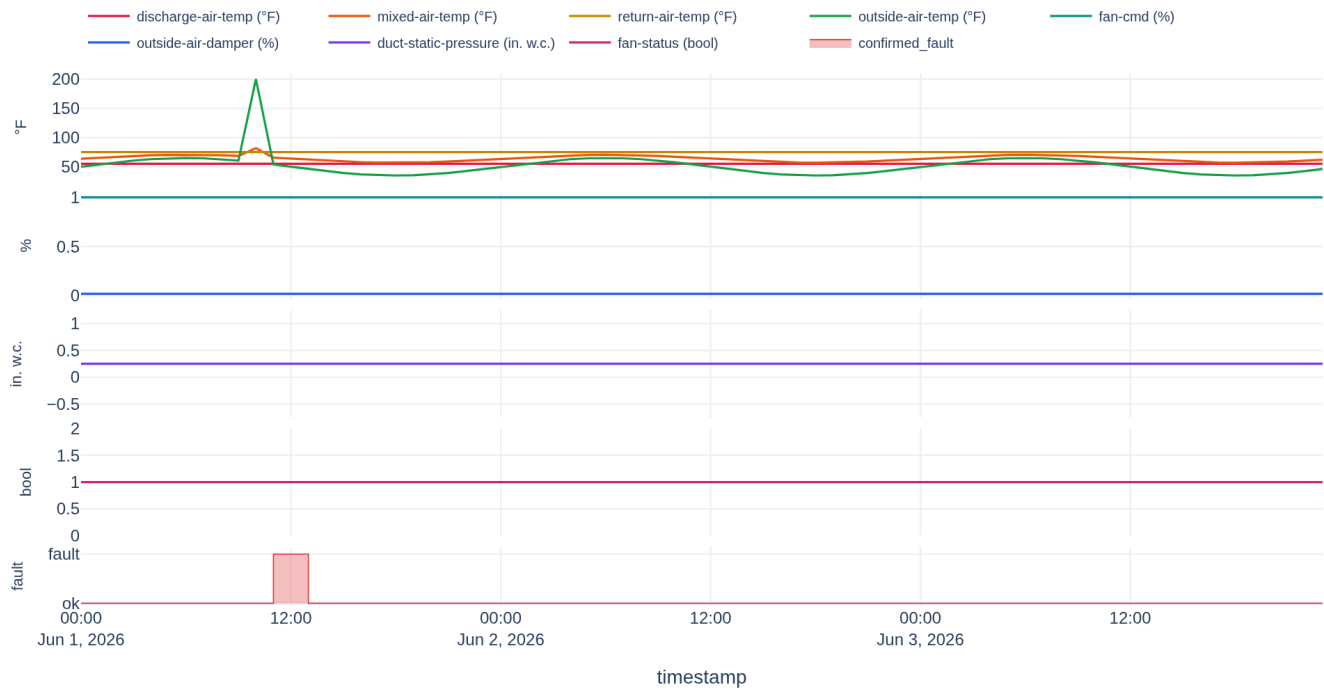
Data: 2.0 fault hours in 2026-06. 2.0h fault (2.8% of active)



SV-RATE – Context-aware sensor rate of change

Rule: Flags implausible sustained sensor rates of change using location- and state-aware thresholds.

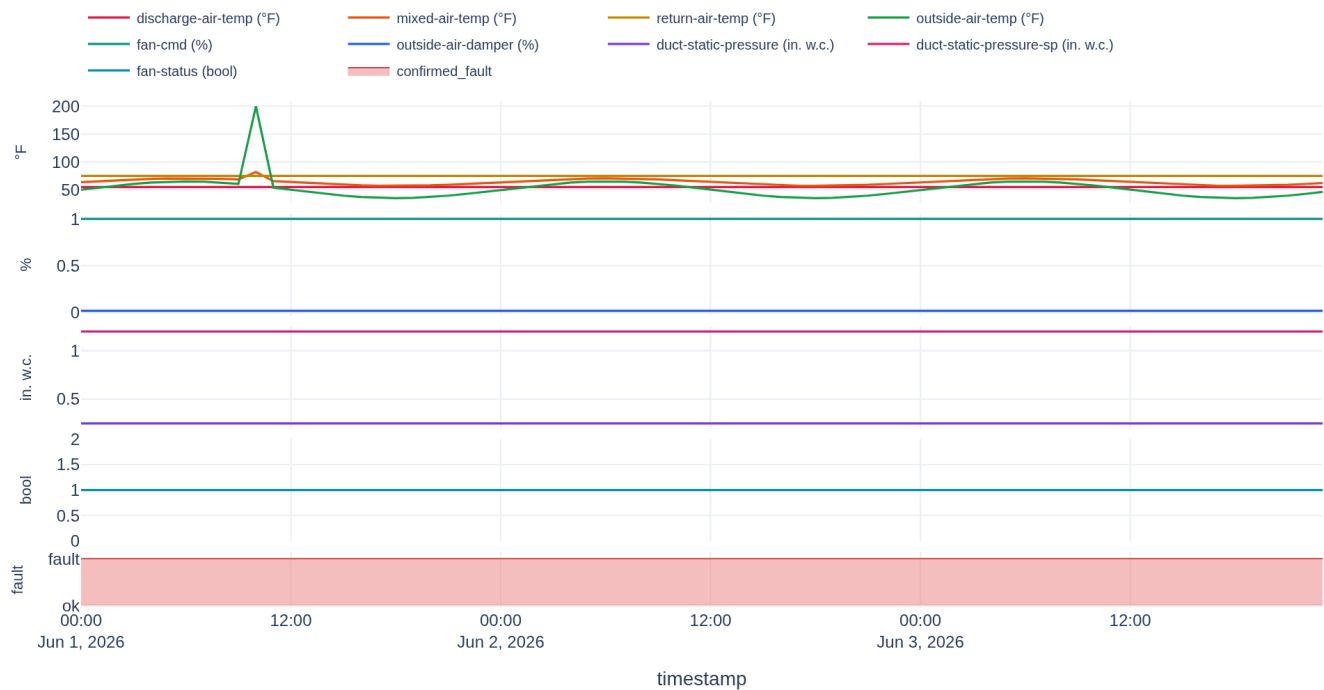
Data: 2.0 fault hours in 2026-06. 2.0h fault (2.8% of active)



FC1 – Duct static below SP at full fan (GL36 A)

Rule: Flags duct static too low while the supply fan is near full speed.

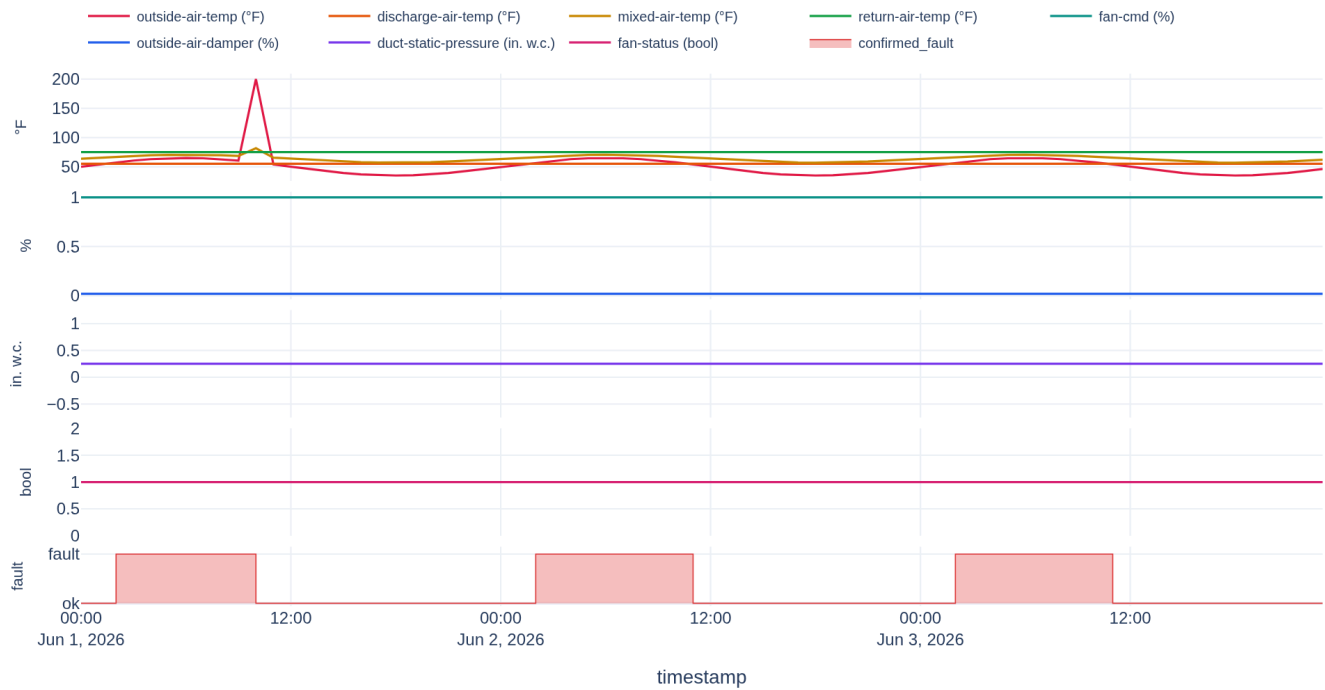
Data: 72.0 fault hours in 2026-06. 72.0h fault (100.0% of active)



ECON-1 – Economizer stuck closed

Rule: Fan on, OA damper < 5%, OAT > 55°F (should be economizing).

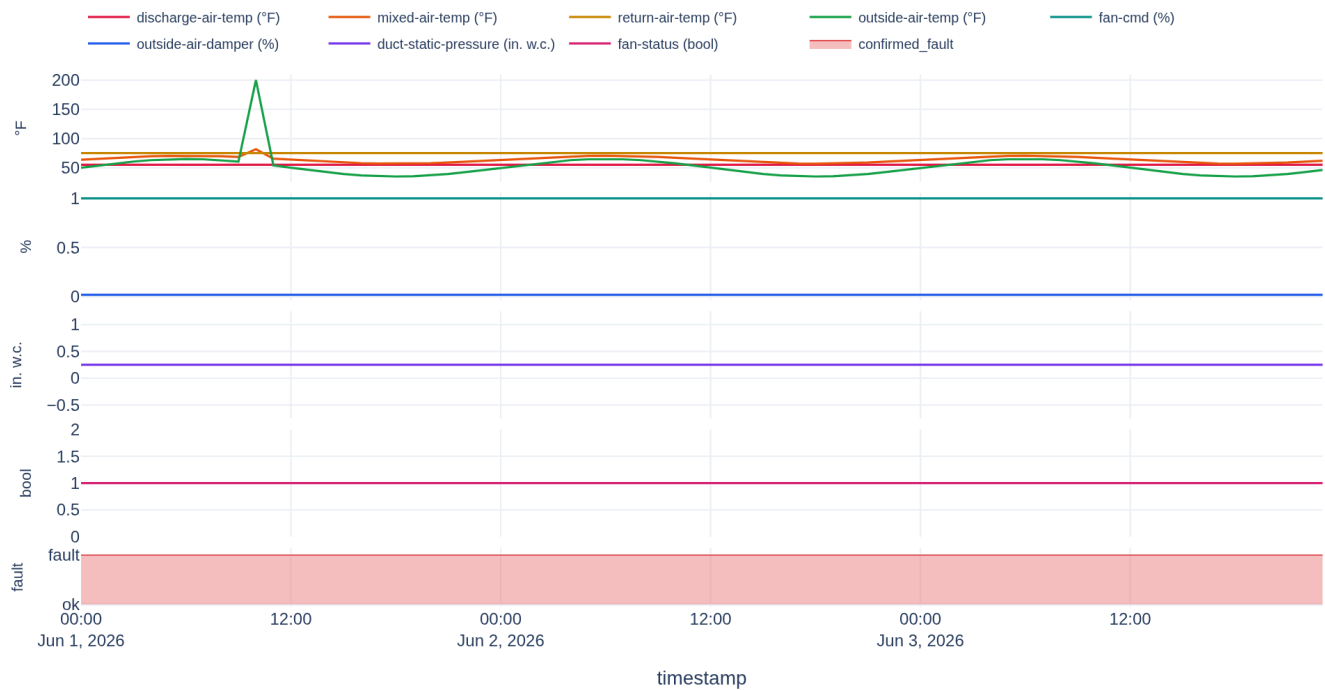
Data: 26.0 fault hours in 2026-06. 26.0h fault (36.1% of active)



SCHED-247 – Always-on fan or pump runtime

Rule: Flags fans or pumps that stay on for nearly the entire analysis window (24/7 runtime).

Data: 72.0 fault hours in 2026-06. 72.0h fault (100.0% of active)



Economizer delta scatter

economizer_delta_scatter: $x = \text{OAT} - \text{RAT} (\text{delta_or_f})$, $y = \text{MAT} - \text{RAT} (\text{delta_mr_f})$. Reference lines $y = \text{OA fraction} * x$ for 0/25/50/75/100% OA. Fan ON and $|\text{OAT}-\text{RAT}| \geq 10 \text{ F}$. Viewport is the

bottom-left mixing quadrant only (both deltas ≤ 0 : $OAT \leq RAT$ and $MAT \leq RAT$). Do not plot $OAT - MAT$ vs $RAT - MAT$.

